

Exam Date & Time: 27-Nov-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

**Motion and Geometry Based Methods in Computer Vision [ECE-4078]**

**Marks: 50**

**Duration: 180 mins.**

### PART-A

**Answer all the questions.**

- 1) Given the following sets of lines, calculate the point of intersection for each pair of lines using homogeneous representation (4)
  - A) a)  $x + 3y = 0$  and  $4x + 5y = 0$ ,  
b)  $x = 4$  and  $x = 7$ .
  - B) Compare the two simple strategies for tracking an object: tracking by detection and tracking by matching. Suggest a possible improvement in the Lukas and Kanade method to compute optical flow in case of significant motion (for instance, motion of more than 50 pixels). (3)
  - C) With the help of a suitable example, explain how dynamic programming could be used to develop an algorithm incorporating ordering constraint to compute corresponding image points  $x, x'$ . (3)
- 2) (4)
  - A)

Calculate the output image  $I_{out}$  obtained by applying 3 x 3 uniform smoothing operator on the 8-bit grayscale image  $I_{in}$  shown in Figure below. Using the output image  $I_{out}$  as an example, identify which operator (Uniform or Gaussian) is a better approximation of a defocused camera. Justify your answer.

|   |   |   |    |    |   |   |   |
|---|---|---|----|----|---|---|---|
| 0 | 0 | 0 | 0  | 0  | 0 | 0 | 0 |
| 0 | 0 | 0 | 0  | 0  | 0 | 0 | 0 |
| 0 | 0 | 0 | 0  | 0  | 0 | 0 | 0 |
| 0 | 0 | 0 | 10 | 10 | 0 | 0 | 0 |
| 0 | 0 | 0 | 10 | 10 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0  | 0  | 0 | 0 | 0 |
| 0 | 0 | 0 | 0  | 0  | 0 | 0 | 0 |
| 0 | 0 | 0 | 0  | 0  | 0 | 0 | 0 |

- B) Given corresponding image points  $x_i, x'_i$  for 3D scene points  $X_i$ . State the appropriate reconstruction needed to identify/measure the following geometric properties (3)
- if a line intersects a plane or not.
  - distance (in cms) between two points. Justify your answer.
- C) Describe the time of flight range sensors. Discuss two limitations of triangulation-based range sensors. (3)

3) (5)

A)

Given two 8-bit grayscale images shown in Figure below, compute the joint probability  $p(x, y)$  of the two images. With the help of these two images as an example, describe the process of image registration using mutual information.

|   |    |    |    |   |
|---|----|----|----|---|
| 0 | 0  | 0  | 0  | 0 |
| 0 | 20 | 20 | 20 | 0 |
| 0 | 20 | 20 | 20 | 0 |
| 0 | 0  | 0  | 0  | 0 |
| 0 | 0  | 0  | 0  | 0 |

**Image 1**

|   |    |    |    |   |
|---|----|----|----|---|
| 0 | 0  | 0  | 0  | 0 |
| 0 | 0  | 0  | 0  | 0 |
| 0 | 20 | 20 | 20 | 0 |
| 0 | 20 | 20 | 20 | 0 |
| 0 | 0  | 0  | 0  | 0 |

**Image 2**

- B) Explain Lukas and Kanade (LK) pseudo-inverse and LK least square method to compute optical flow. Compare LK method with Horn and Schunck's approach to compute optical flow. (5)
- 4) Given two uncalibrated cameras, explain the steps to compute an affine reconstruction of the scene from image correspondence. (4)
- A)
- B) Explain why a RANSAC based approach cannot be used to register deformable objects. Describe at least two applications that require registering deformable objects. (3)
- C) Describe range images. Explain the utility of range images in 3D reconstruction. (3)
- 5) Differentiate between stratified approach and direct approach for upgrading a projective reconstruction to metric reconstruction. State and describe the geometric primitives needed in both these approaches for upgrading a projective reconstruction to metric. (5)
- A)
- B) Explain bundle adjustment-based reconstruction method for uncalibrated perspective camera. Discuss its limitations and advantages. (5)

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